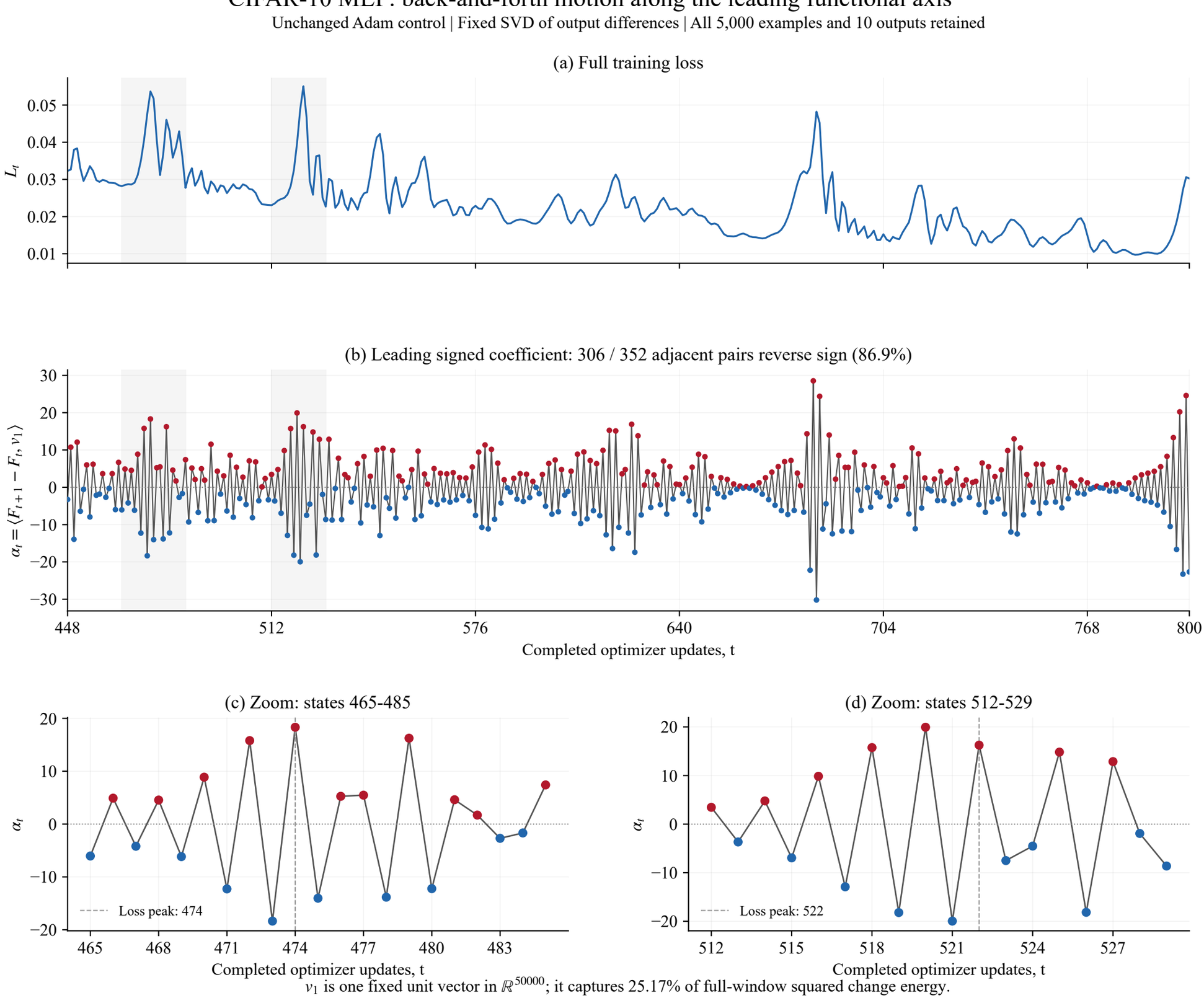

CIFAR-10 MLP: back-and-forth motion along the leading functional axis
Unchanged Adam control | Fixed SVD of output differences | All 5,000 examples and 10 outputs retained
(a) Full training loss
L_t
0.05
0.04
0.03
0.02
0.01
(b) Leading signed coefficient: 306 / 352 adjacent pairs reverse sign (86.9%)
$\alpha_t = \langle F_{t+1} - F_t, v_1 \rangle$
30
20
10
0
−10
−20
−30
448
512
576
640
704
768
800
Completed optimizer updates, t
(c) Zoom: states 465-485
α_t
20
10
0
−10
−20
Loss peak: 474
465
468
471
474
477
480
483
Completed optimizer updates, t
(d) Zoom: states 512-529
α_t
20
10
0
−10
−20
Loss peak: 522
512
515
518
521
524
527
Completed optimizer updates, t
v_1 is one fixed unit vector in $\mathbb{R}^{50000}$; it captures 25.17% of full-window squared change energy.
Raw coefficients shown (heatmap row 1 multiplied by $\sqrt{5000}$). Positive and negative mean opposite motion.
α_t describes update $t \to t+1$; L_t is the loss before that update. Basis fitted over the entire window.